

Verify motor direction in Betaflight Configurator (Motors tab)
Continuity check all power rails before plugging battery
Power MANDATORY before first power | Props OFF for motor test

=== SAFETY ===

Wiring: CAM to VTX | JST-GH 1.25mm: GPS, I2C | Heat shrink all joints
Power: GND, 5V rail | 26AWG: UART, ESC signals | 28AWG: I2C, sensors

=== WIRING ===

Prop: Under 250g (limit) | CG: <1mm offset | ALL DIRECT FIT – NO MODS
Prop: 3052 3in tri-blade | 5mm hub+1.5mm adapter | 22.8mm clearance
Mount: NUT(2)+FC(6.5)+NUT(2)=15mm | Standoff: 25mm | Clearance: 10mm
Frame: 10mm 4mm carbon | Twin 20x20 tower | 8x M3x25mm alu standoffs

=== MECHANICAL ===

Power: 1S LiHV (7.6V 75C) XT30 | 30x18x15mm | 4-5min flight | LVC 3.5V/cell
Tx: 1in1 20x20 | 3-6S | DShot600 | 25x25x4.5mm | Bidirectional ON
Rx: 1405 Mini 20x20 | ICM42688 | DPS310 | 25x25x6.5mm | 4 UARTs

=== COMPONENT SPECS ===

Motor: VTX->RX1 >=25mm | RX1->RX2 >=40mm | All >10mm from carbon/props
Mount: 1ft standoff (60mm active) | ANT2: Horizontal top plate rear (60mm)
Radio: 1RD ELRS Diversity | 22x14mm 1.1g | Dual SX1280 | CRSF at 420000

=== RECEIVER ===

Power: 350mW 16x16->20x20 | SMA -> Foxeer Lollipop RHCP (rear, 50mm)
Video: 1200TVL 14x14->19x19 | VID -> VTX CAM_IN (JST-SH 1.0mm 4-pin)

=== CAMERA and VTX ===

Motor: 1804 3500KV | 13.3g | 12x12 M2 | 22.68x13.2mm | 1.5mm shaft
ESC: C S3 -> Motor RL (CCW) | M4: FC M4 -> ESC S4 -> Motor RR (CW)
ESC: C S1 -> Motor FL (CCW) | M2: FC M2 -> ESC S2 -> Motor FR (CW)

MOTOR OUTPUTS (DShot600) ===

GPS: 13): FC <=> GPS (optional) | UBLOX | 57600 baud | M100-5883
Radio: 1(TX5/RX5): FC <=> RX CRSF | ELRS V3 | 420000 baud
Audio: 1: FC -> VTX SmartAudio RX | SmartAudio 2.1 | 115200 baud

=== SIGNAL ROUTING ===

Power: 1-> VTX(Zeus350) + CAM(CaddxAnt) + RX(SuperD) + GPS(M100-opt)
Battery: 1mAh LiHV (7.4V) XT30 -> FC VBAT -> Internal 5V/3A BEC

=== POWER ===

ANALOG FPV – ELECTRICAL SCHEMATIC

GROK – FPV Drone Engineering

Sheet: /
File: gx140_clean.kicad_sch

Title: GX140 3in Analog FPV – Electrical Schematic

Size: A3	Date: 2025-12-12	Rev: 1.0
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1